

Do Machine Learning Force Fields Overcome Adversarial Attacks?

Declan Chan — Engineering Science 2T9, U of T
AutoDIAL Research Group
Department of Materials Science & Engineering
UnERD 2026



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Why Do We Use Machine Learning Force Fields?

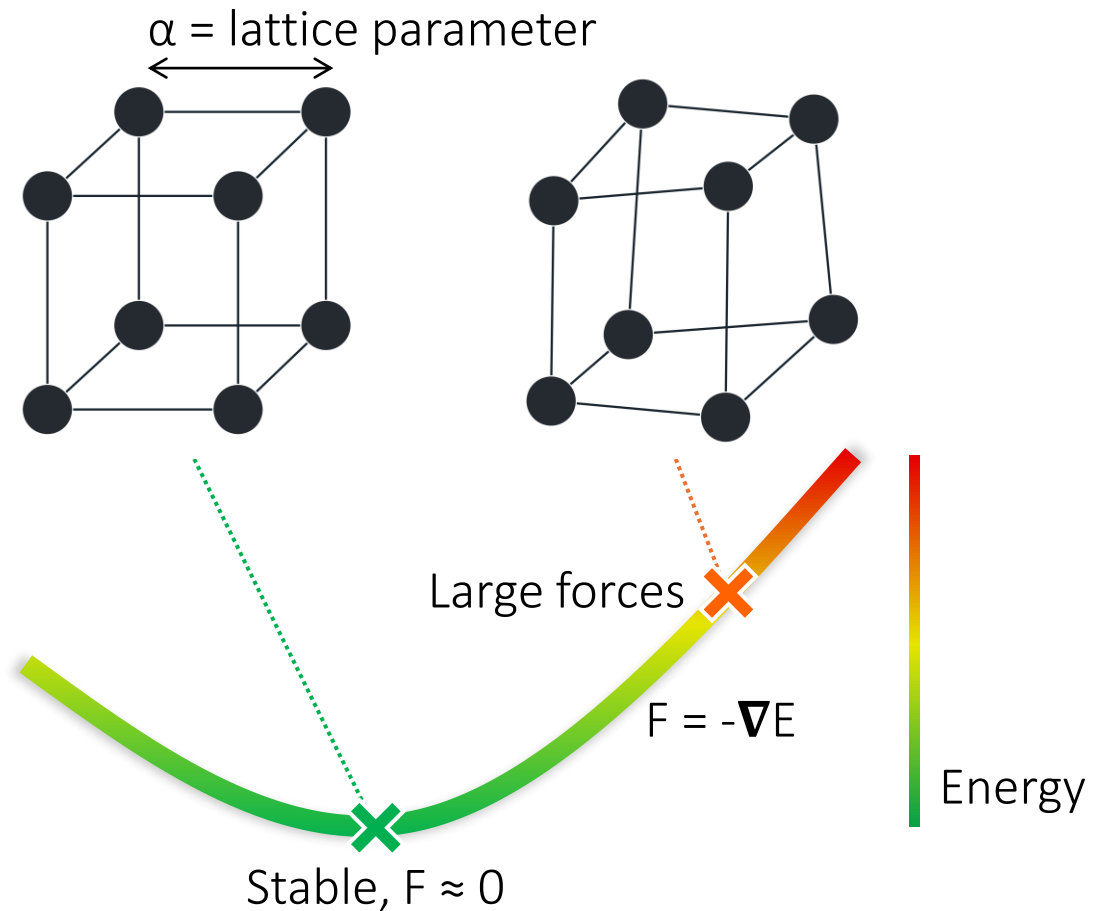
Screen many atomic structures for materials discovery

→ **D**ensity **F**unctional **T**heory

To save time, cost & resources

→ **M**achine **L**earning **F**orce **F**ields

MLFFs learn the DFT PES to predict energies & forces efficiently.



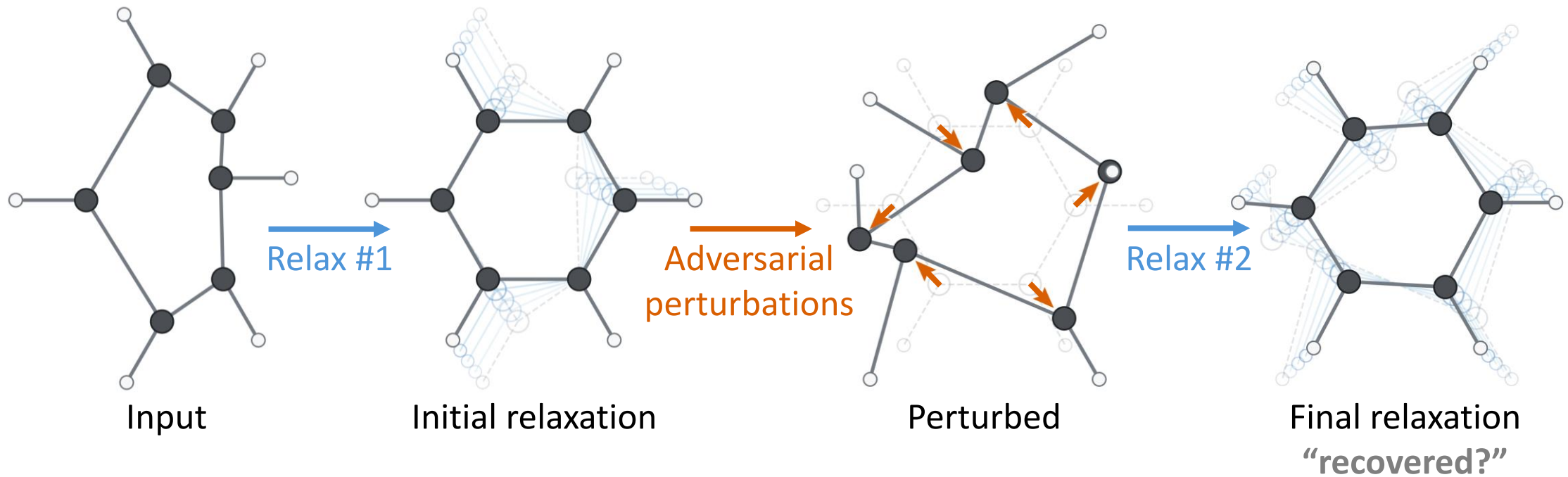
What Happens When MLFFs Leave Their Comfort Zone?

Risk → Out-of-distribution structures may lead to unreliable predictions.

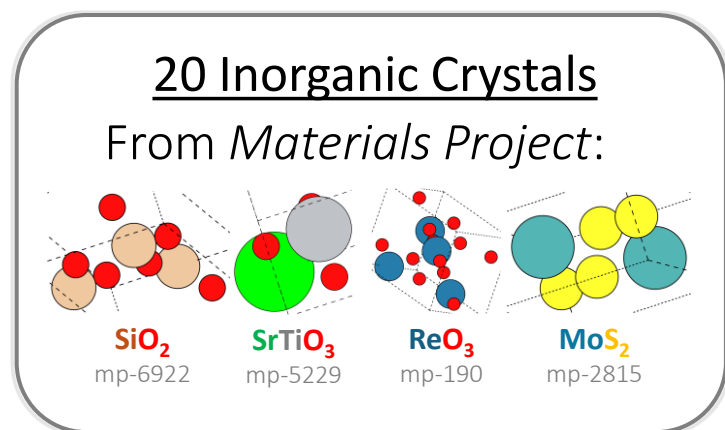
Conventional benchmarks use static errors against DFT but miss:
unphysical artifacts and structural stability.

Assessing MLFF Robustness Using Perturbations

How do MLFFs relax structures from adversarial perturbations?



Experimental Design: 20 Structures and 2 MLFFs

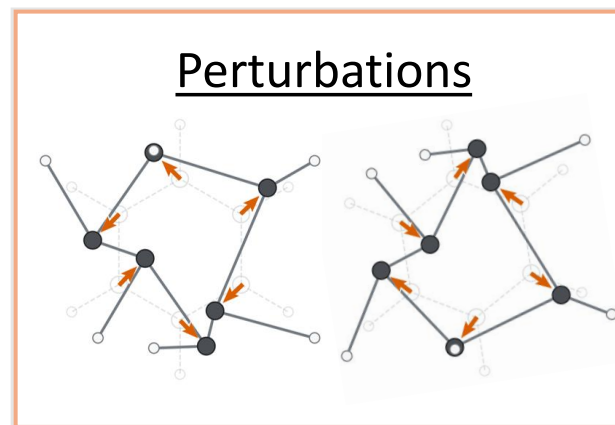


Initial
relaxation ↓

MLFFs

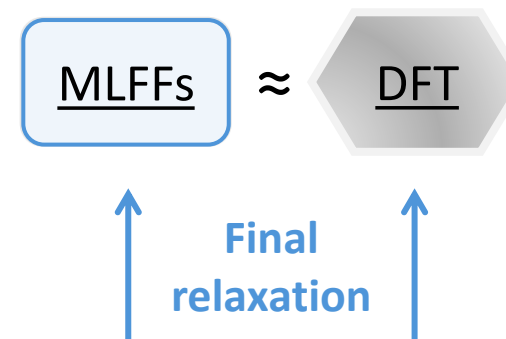
Model 1: UMA—universal models for atoms
Model 2: MACE-MH—multi-head message passing

Perturb
structures →

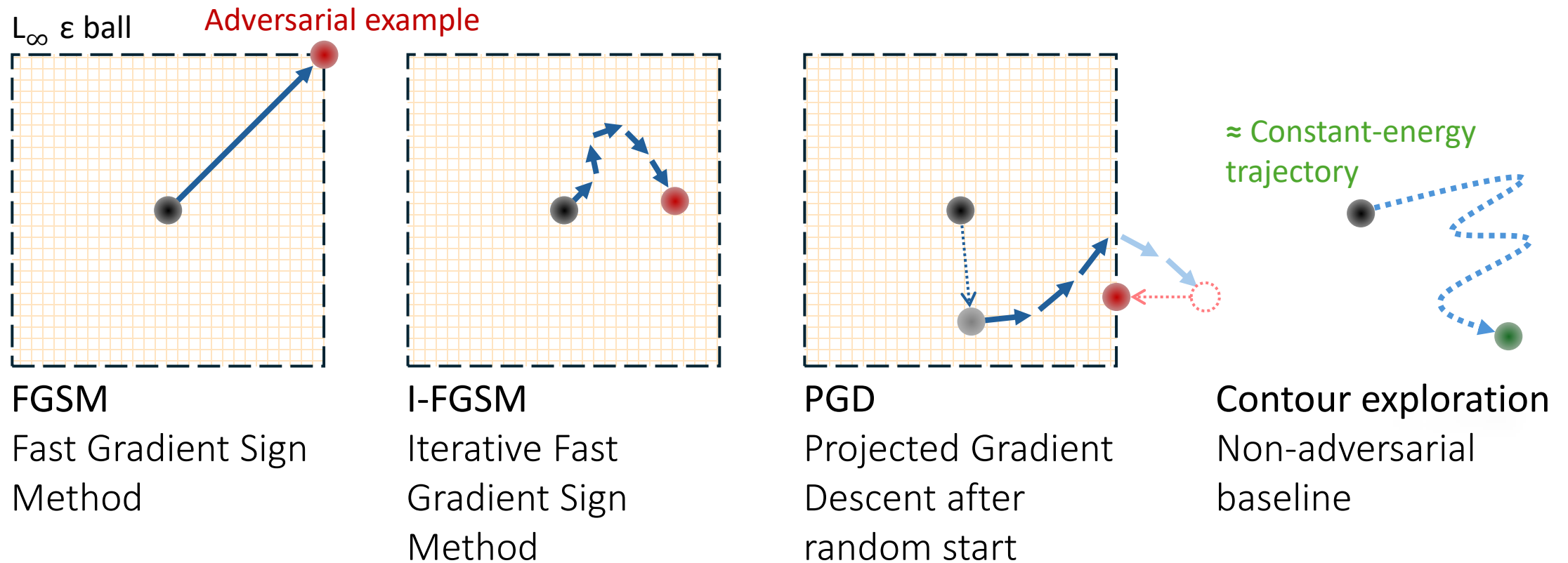


Final recovered structure = initial structure?

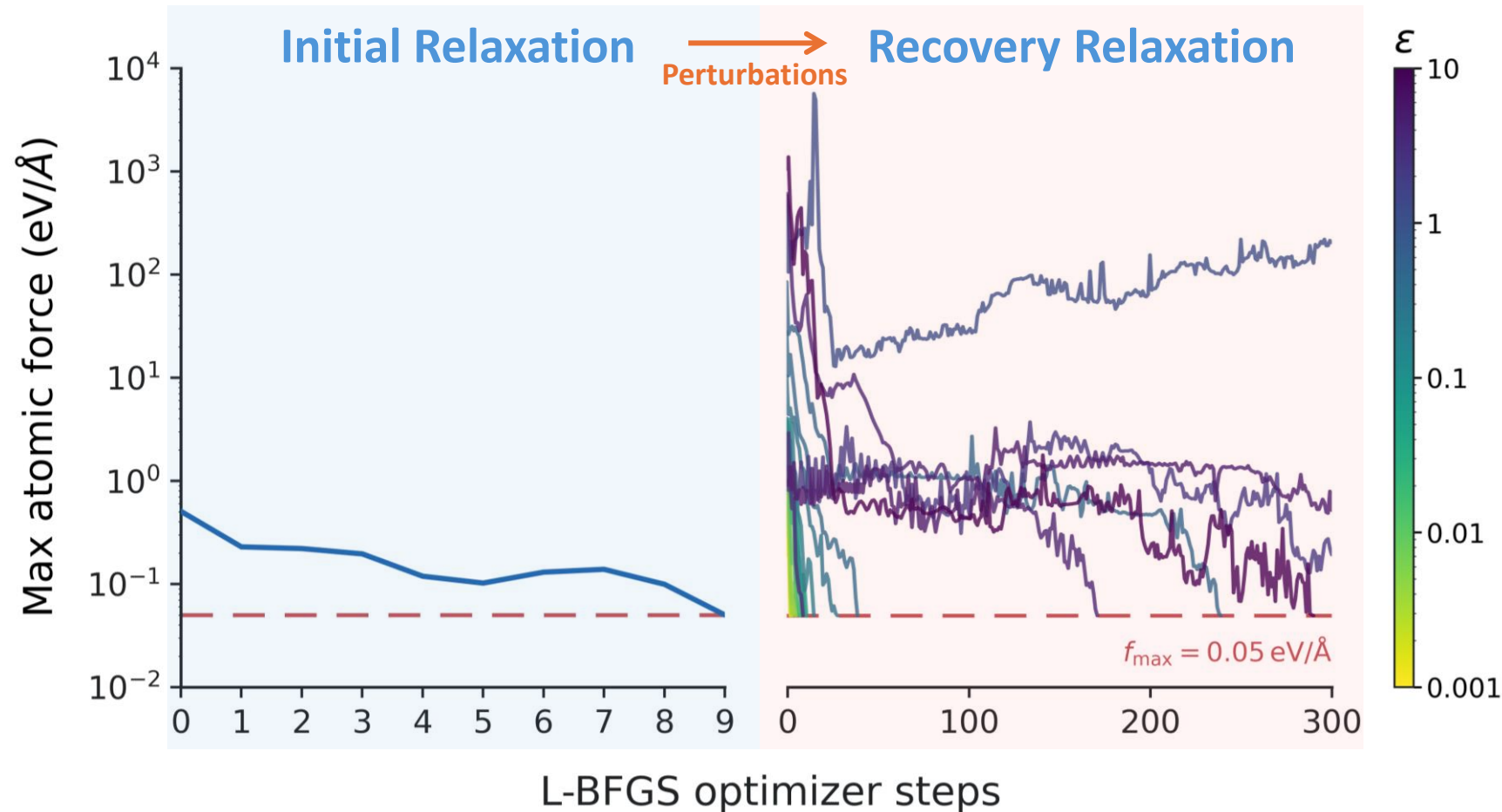
Measure changes in energy, forces, topology



Robustness Tests Through Adversarial Perturbations and Contour Exploration

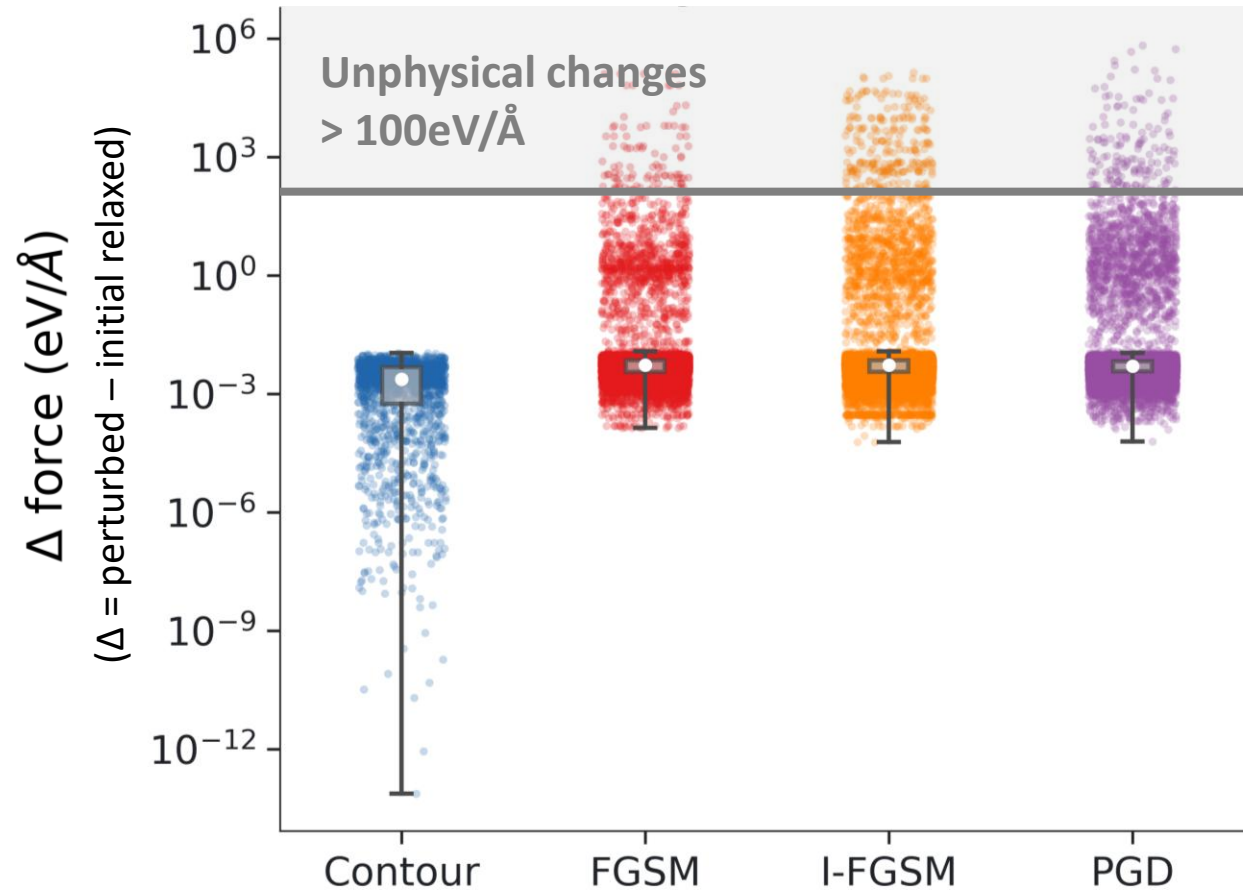


Testing MLFF Relaxation Recovery After Perturbations



Experiments repeated across:	
dtype	Seed
float32	42
	43
	44
float64	45
	46

Adversarial Gradients Expose Unphysical MLFF Predictions



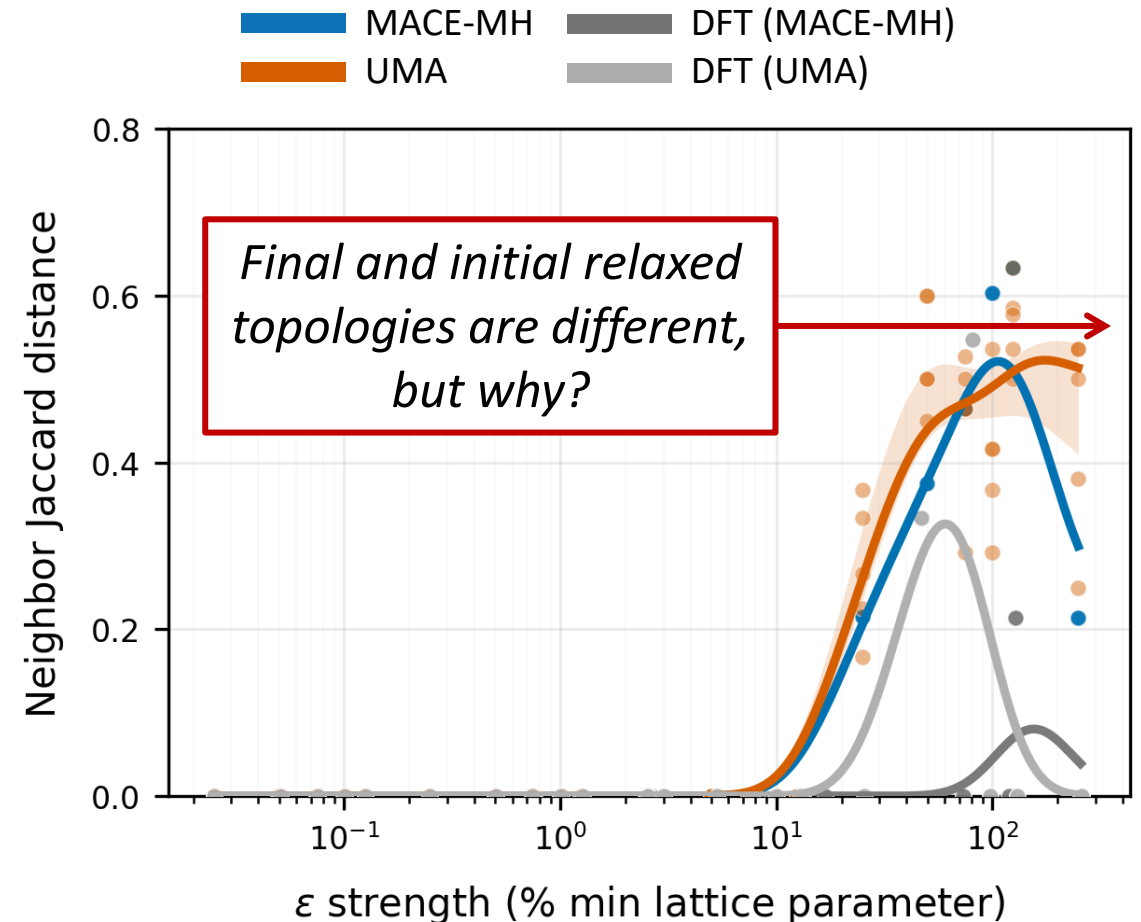
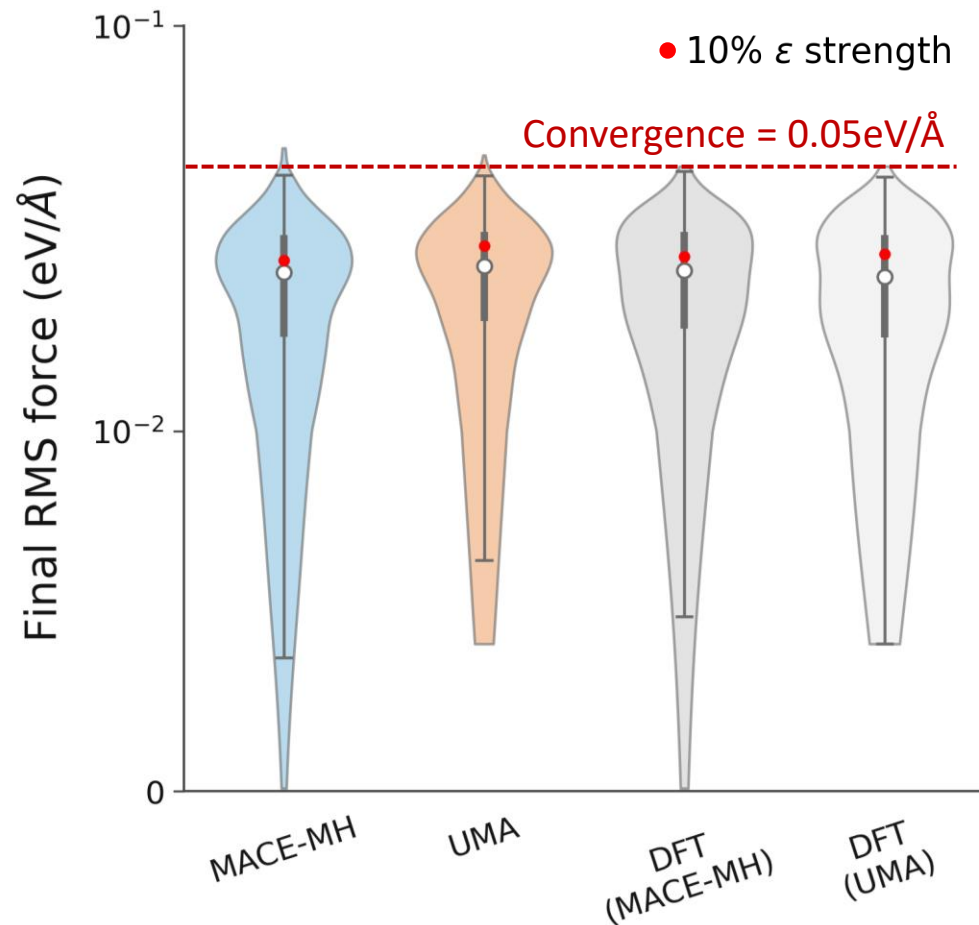
Convergence rate:

94% adversarial

100% non-adversarial

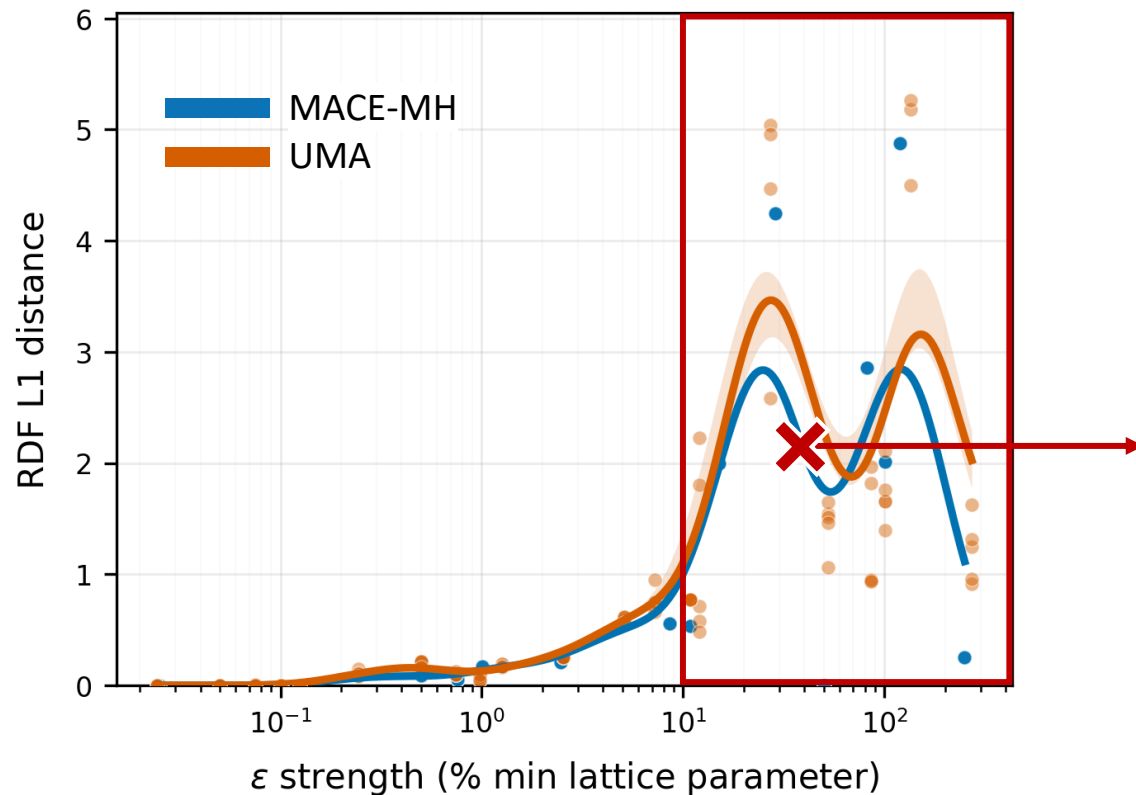
*After perturbations,
relaxations can converge to
low forces—but do they
recover structurally?*

Although Forces Converge, Topology Fails to Recover $> 10\% \epsilon$

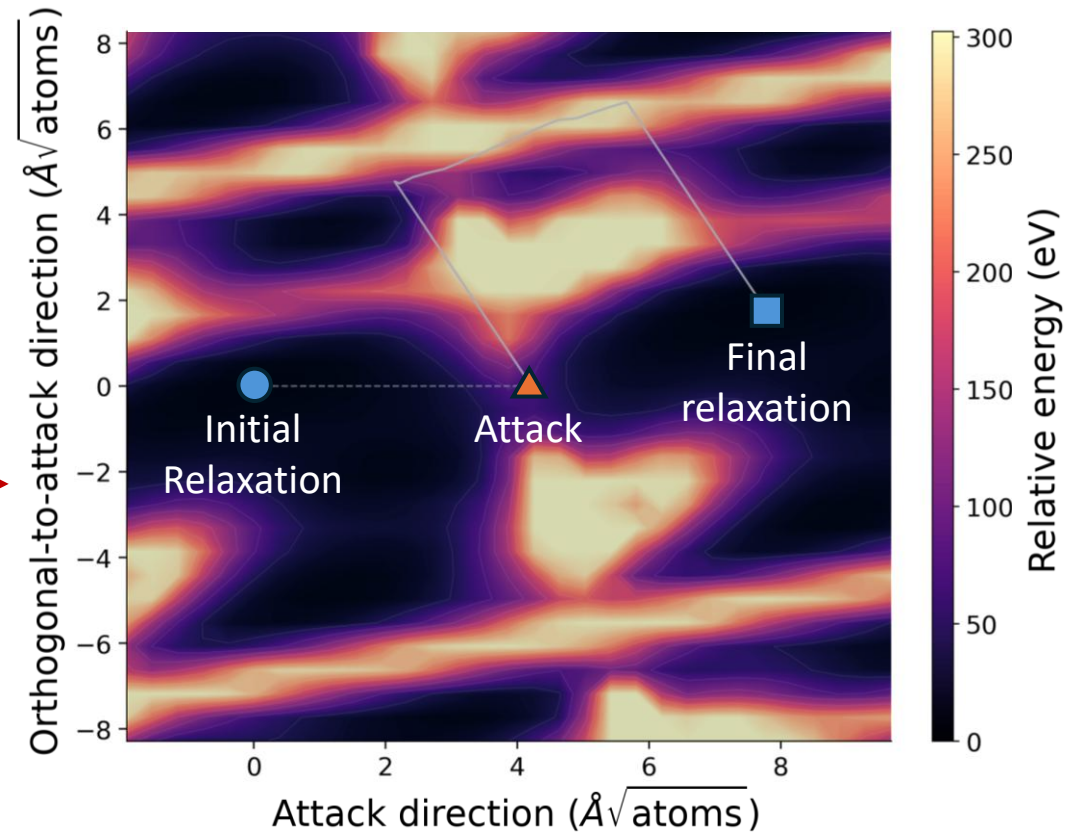


Structural Recovery Fails Because MLFFs Relax into Different Basins

Geometric Differences in MLFF Recovery and DFT



Example MLFF PES for $\epsilon > 10\%$ (hBN)



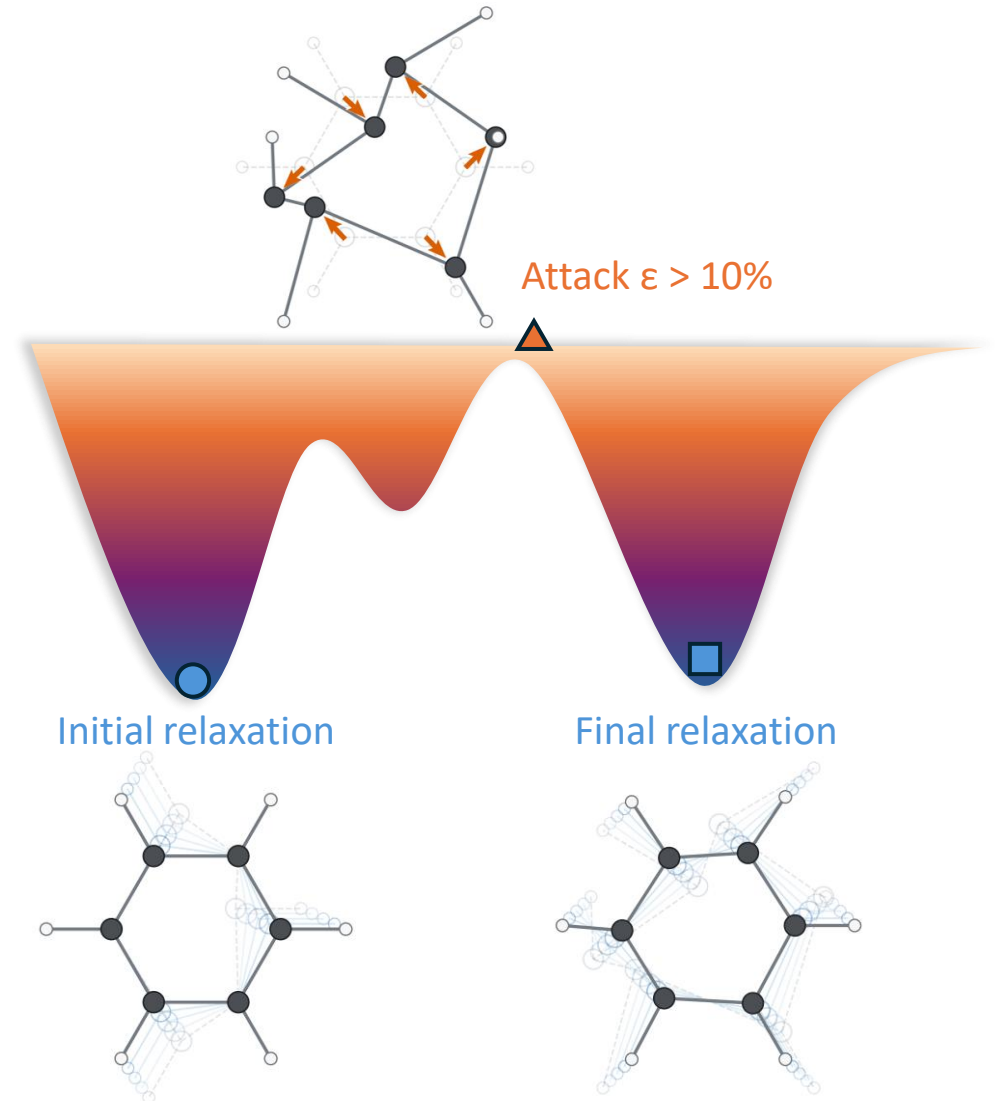
Towards More Reliable MLFFs for Materials Discovery

Takeaways:

- Low forces $\neq$ structural recovery
- Adversarial gradients redirect relaxations
- 10% perturbation threshold observed in the dataset.

Ongoing:

- More MLFFs & structures
- Dynamic stability with phonons



Thank You

Acknowledgements

- Professor Jason Hattrick-Simpers
- Dr. Ashley Dale, Dr. Dan McHaffie, and Dr. Hao Wan
- AutoDIAL Research Group
- Department of Materials Science & Engineering
- Digital Research Alliance of Canada



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